Quantum Implicit Q-Learning (Q-IQL): Finalized Architecture Proposal

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Abstract

We present the finalized architecture for Quantum Implicit Q-Learning (Q-IQL), a hybrid quantum–classical offline reinforcement learning agent. Q-IQL replaces the state-value network $V_\psi(s)$ of Kostrikov et al.'s Implicit Q-Learning (IQL) [1] with an 8-qubit Data Re-uploading (DRU) variational circuit [5], while keeping the twin critic $Q_\theta(s, a)$ and actor $\pi_\phi(a \mid s)$ classical. The circuit uses local Pauli-Z readout and identity-block initialisation [6] to operate within the trainable regime guaranteed by Cerezo et al. [3]. Primary development targets classical simulation via PennyLane [9]; Phase 2 extends to the Quantinuum H-series (98 qubits).

1. Introduction

Offline RL trains agents from fixed datasets without environment interaction [13]. IQL [1] achieves state-of-the-art results by decomposing policy optimisation into three supervised losses — value expectile regression, Bellman critic regression, and advantage-weighted behavioural cloning — without ever querying out-of-distribution actions.

Variational quantum circuits (VQCs) have been proposed as quantum-enhanced function approximators, motivated by exponential parameter efficiency arguments [14]. However, VQCs suffer from *barren plateaus* (BPs) [2]: exponentially vanishing gradients that make gradient-based training intractable. This risk is compounded in offline RL, where sparse reward signals and a fixed dataset prevent any corrective data collection.

Q-IQL targets the intersection of these domains. The key architectural contributions are: (i) quantising only $V_\psi(s)$, the network with the most leverage on overall training quality and the simplest input–output structure; (ii) adopting the DRU ansatz [5] with a six-point BP mitigation stack grounded in Cerezo et al. [3] and Grant et al. [6]; (iii) providing a clear path to Phase 2 on Quantinuum 98-qubit hardware.

2. Background

2.1. IQL Training Objectives

IQL maintains three networks — $V_\psi(s)$, $Q_\theta(s, a)$, $\pi_\phi(a \mid s)$ — updated via:

$$\mathcal{L}_V(\psi) = \mathbb{E}_{(s,a)\sim\mathcal{D}}[\ell_\tau(\mathrm{sg}(Q_\theta(s,a)) - V_\psi(s))]\,, \quad \ell_\tau(u) = |\tau - \mathbf{1}[u < 0]| \cdot u^2, \tag{1}$$

$$\mathcal{L}_Q(\theta) = \mathbb{E}\left[\sum_{i=1}^{2}\left(Q_\theta^{(i)}(s,a) - r - \gamma \bar{V}_\psi(s')\right)^2\right], \tag{2}$$

$$\mathcal{L}_\pi(\phi) = -\mathbb{E}[\exp(\beta\,(\mathrm{sg}(Q_\theta) - \mathrm{sg}(V_\psi)))\log \pi_\phi(a \mid s)]\,, \tag{3}$$

where ℓ_τ is the asymmetric expectile loss at level τ, $\bar{V}_\psi$ is a Polyak-averaged target of V_ψ, and $\mathrm{sg}(\cdot)$ denotes a stop-gradient.

2.2. Barren Plateau Theory

Theorem 1 (McClean et al., 2018 [2]). *For random parameterised circuits approximating unitary 2-designs,* $\mathrm{Var}[\partial_k C] \sim \mathcal{O}(2^{-n})$.

Theorem 2 (Cerezo et al., 2021 [3]). *For an alternating layered ansatz: (i) a* global *observable induces exponential gradient vanishing at* any *depth; (ii) a* local *k-qubit observable with $L \in \mathcal{O}(\log n)$ gives* $\mathrm{Var}[\partial_k C] \in \Omega(1/\mathrm{poly}(n))$.

Theorem 2(ii) is the primary design constraint: *local observables and shallow circuits are both necessary* to stay in the polynomially-vanishing regime. Additional BP sources relevant to Q-IQL include noise-induced BPs $(1-\lambda)^L$ [10], entanglement-induced BPs for volume-law entanglement [11], and the impossibility of escaping BPs via gradient-free optimisers [12].

3. Architecture

3.1. Component Decisions

Table 1: Q-IQL component decisions. Phase 1: PennyLane simulation. Phase 2: Quantinuum H-series (98 qubits).

Component	Phase 1	Phase 2	Rationale
$V_\psi(s)$	**Quantum**	Quantum (larger)	Scalar output; state-only input (11 dims); feeds Bellman backup $\bar{V}_\psi$, affecting all three losses.
$Q_\theta(s,a)$	Classical	**Quantum**	Joint input (14 dims) needs $\geq$12 qubits; feasible on Quantinuum; zero BP risk when classical.
$\pi_\phi(a \mid s)$	Classical	Classical	Continuous Gaussian output with tanh squashing is not naturally expressible via qubit measurements.
$\bar{V}_\psi$ (target)	EMA of V_ψ	EMA of V_ψ	Polyak update applies to quantum parameters identically.

3.2. Why the Value Network, Not the Actor

The actor $\pi_\phi(a \mid s)$ ultimately selects actions at deployment, but in IQL it is trained via advantage-weighted behavioural cloning (Eq. (3)). The advantage $A(s,a) = Q_\theta(s,a) - V_\psi(s)$ is computed from V_ψ, so a more accurate quantum V_ψ directly sharpens the actor's training signal without requiring the actor to be quantum. Furthermore, V_ψ also enters the Bellman backup target $\bar{V}_\psi(s')$ in Eq. (2), meaning a better quantum V_ψ simultaneously improves critic and actor training — a multiplier effect absent if only the actor were quantised. Generating a well-calibrated squashed Gaussian from qubit measurements would require ancilla sampling procedures not supported by standard VQC frameworks.

4. Quantum Circuit Specification

4.1. Preprocessing and Encoding

The raw observation $s \in \mathbb{R}^{11}$ is preprocessed as:

$$\mathbf{x}_s = \arctan\left(\frac{s - \mu_{\mathcal{D}}}{\sigma_{\mathcal{D}}}\right) \in \left(-\tfrac{\pi}{2}, \tfrac{\pi}{2}\right)^{11}, \tag{4}$$

bounding angle encoding inputs without hard clipping. The 11-dimensional vector is distributed across DRU layers via zero-padding to width 8.

4.2. Data Re-uploading Ansatz

We adopt the multi-qubit DRU architecture of Pérez-Salinas et al. [5], the recommended ansatz from the Week 1 BP analysis. Each layer $L^{(i)}$ combines trainable parameters with data re-uploading:

$$L^{(i)} = B_{\mathrm{CZ}} \cdot \bigotimes_{j=1}^{n} R_j\left(\boldsymbol{\theta}_j^{(i)} + \mathbf{w}_j^{(i)} \odot \mathbf{x}_s\right), \tag{5}$$

where $R_j(\cdot) = R_z R_x R_z$ is an SU(2) rotation on qubit j, $\boldsymbol{\theta}_j^{(i)} \in \mathbb{R}^3$ are trainable bias angles, $\mathbf{w}_j^{(i)} \in \mathbb{R}^3$ are trainable data weights, and B_{CZ} is a fixed nearest-neighbour CZ entangling layer. The full circuit is:

$$U(\boldsymbol{\theta}, \mathbf{w}, \mathbf{x}_s) = L^{(3)} \cdot L^{(2)} \cdot L^{(1)} \cdot B_{\mathrm{pre}}, \tag{6}$$

where B_{pre} is a fixed non-trained CZ preamble generating initial entanglement without contributing to the trainable BP depth [6].

The readout is a local Pauli-Z expectation on qubit 1, satisfying Theorem 2(ii):

$$\hat{V}(s) = \left\langle \sigma_z^{(1)} \right\rangle_{U(\boldsymbol{\theta}, \mathbf{w}, \mathbf{x}_s)}, \qquad V_\psi(s) = a\,\hat{V}(s) + b, \tag{7}$$

where $a, b \in \mathbb{R}$ are a trainable affine output head scaling the measurement from $[-1, 1]$ to the full return range.

4.3. Parameter Count

Table 2: Parameter count comparison: classical vs. quantum V_ψ.

	Classical MLP	**Phase 1** (n=8)	**Phase 2** (n=16)
Architecture	[256,256]	8-qubit DRU, $L = 3$	16-qubit DRU, $L = 4$
Variational params	—	$6 \times 8 \times 3 = 144$	$6 \times 16 \times 4 = 384$
Output head params	$\approx 68{,}000$	2 (affine)	2 (affine)
Total	**68,096**	**146**	**386**

The 466× parameter reduction motivates the hypothesis that quantum function approximation may generalise more effectively from the fixed offline dataset. Gradients are computed via the parameter-shift rule [8]: $\partial_{\theta_k} C = [C(\theta_k + \pi/2) - C(\theta_k - \pi/2)]/2$, requiring $2 \times 144 = 288$ circuit evaluations per backward pass in Phase 1.

5. Barren Plateau Mitigations

Three Q-IQL-specific risk factors compound the standard BP problem: (1) **offline data** — no corrective data collection when gradients vanish; (2) **sparse rewards** — reward density $f \ll 1$ dilutes gradients by factor f; (3) **expectile asymmetry** — the factor $|\tau - \mathbf{1}[u<0]|$ further suppresses or amplifies the already small VQA gradient. Note that Q_θ is classical and entirely BP-free; risk is confined to V_ψ.

We adopt the following mitigation stack:

1. **Local observable.** Use $O_V = \sigma_z^{(1)}$; never a global Pauli string $\bigotimes_j \sigma_z^{(j)}$ (Theorem 2(i)).
2. **Identity-block init** [6]. Set $\boldsymbol{\theta}_j^{(i)} \sim \mathcal{U}(0, 2\pi)$ and $\mathbf{w}_j^{(i)} = -\boldsymbol{\theta}_j^{(i)}$, so each layer evaluates to the identity at $t = 0$ for any $\mathbf{x}_s$, guaranteeing non-vanishing initial gradients.

3. **Fixed entangling preamble.** B_{pre} introduces entanglement without adding to trainable BP depth [6].
4. **Shallow circuit.** Keep $L \leq \lfloor \log_2 n \rfloor$ (for $n = 8$: $L \leq 3$), maintaining the polynomial guarantee of Theorem 2(ii).
5. **Layer-wise training** [7]. Begin with $L = 1$ and grow: steps 0–100k ($L = 1$), 100–300k ($L = 2$), 300k–1M ($L = 3$).
6. **Faithful local cost.** Verify empirically that $\left\langle \sigma_z^{(1)} \right\rangle \to 0$ at the expectile loss minimum [3].

6. Training Protocol and Hyperparameters

Algorithm 1: Q-IQL training step

Input: Batch $\{(s, a, r, s', d)\}^B \sim \mathcal{D}$

`// Value step - quantum backward via parameter-shift`

$\mathbf{x}_s \leftarrow \arctan((s-\mu_{\mathcal{D}})/\sigma_{\mathcal{D}})$; $\quad V_\psi(s) \leftarrow a \left\langle \sigma_z^{(1)} \right\rangle_{U(\boldsymbol{\theta}, \mathbf{w}, \mathbf{x}_s)} + b$;

Minimise $\mathcal{L}_V$ (Eq. 1); update $\boldsymbol{\theta}, \mathbf{w}$ (param.-shift), a, b (autograd);

`// Critic step - classical autograd`

$\bar{V}_\psi(s') \leftarrow$ EMA forward pass (`no_grad`); $\quad$ minimise $\mathcal{L}_Q$ (Eq. 2); update θ;

`// Actor step (post-warmup) - classical autograd`

$A \leftarrow \min(\exp(\beta(\text{sg}(Q_\theta) - \text{sg}(V_\psi))), c_{\max})$; $\quad$ minimise $\mathcal{L}_\pi$ (Eq. 3); update ϕ;

`// Target update - Polyak on all params incl.` $\boldsymbol{\theta}, \mathbf{w}$

$\bar{\psi} \leftarrow (1-\rho)\bar{\psi} + \rho\psi$;

Table 3: Q-IQL hyperparameters for Hopper-v4 / mujoco/hopper/medium-v0.

Hyperparameter	Value	Type	Source
Expectile τ	0.7 (0.9 AntMaze)	Shared	[1]
Inv. temp. β	3.0	Shared	[1]
Discount γ	0.99	Shared	[1]
Polyak ρ	0.005	Shared	[1]
Batch size	256	Shared	[1]
Adv. clip $c_{\max}$	100	Shared	[1]
Total steps	1 000 000	Shared	—
Classical hidden dims	[256,256]	Classical	[1]
Classical lr	3×10^{-4}	Classical	[1]
Qubits n	8	Quantum	This work
DRU layers L	3	Quantum	This work
Entanglement	CZ ladder	Quantum	[3]
Observable	$\sigma_z^{(1)}$	Quantum	[3]
Init. $\mathbf{w}^{(i)}$	$-\boldsymbol{\theta}^{(i)}$	Quantum	[6]
Differentiation	parameter-shift	Quantum	[8]
Quantum lr	1×10^{-3}	Quantum	Tunable
Layer schedule	L=1: 0–100k; L=2: 100–300k; L=3: 300k+	Quantum	[7]
Backend (Phase 1)	`lightning.qubit`	Quantum	[9]

7. Conclusion

Q-IQL replaces the classical state-value network $V_\psi(s)$ with an 8-qubit Data Re-uploading circuit, leaving $Q_\theta(s, a)$ and $\pi_\phi(a \mid s)$ classical. This boundary is motivated by three converging arguments: V_ψ is the central computational object in IQL (feeding both critic and actor); its scalar, state-only structure minimises qubit encoding requirements; and the BP theory of Cerezo et al. [3] provides a concrete trainability guarantee for shallow DRU circuits with local observables. The six-point mitigation stack addresses all identified risk factors. Phase 2 extends to quantum $Q_\theta(s, a)$ on Quantinuum's 98-qubit H-series, targeting a fully quantum hybrid forward pass.

Codebase mapping. Only two files require modification from the classical baseline: `networks.py` (add `QuantumValueNetwork` via `qml.qnn.TorchLayer`) and `trainer.py` (add `QuantumIQLTrainer` overriding `_build_networks()`). All losses, replay buffer, evaluation, and checkpointing are inherited unchanged.

References

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